

Final Project — STM32 Manufacturing Test System

Final verification report

The system implements PC-to-STM32 manufacturing verification over Ethernet/UDP. The STM32 parses commands, executes TIMER, UART, SPI, I2C or ADC tests, and returns Success/Failure.

Requirements coverage

Requirement	Status
Ethernet / UDP / protocol	PASS
TIMER	PASS
UART + DMA	PASS
UART CRC32 >100 bytes	PASS
SPI + DMA	PASS
I2C + DMA/IRQ	PASS
ADC 12-bit	PASS
C++17 Linux CLI	PASS
SQLite persistence	PASS
Edge cases	PASS
Doxygen coverage	REVIEW
Board note (F746 assignment / F756 implementation)	DOCUMENT

UART CRC32 verification

A 128-byte UART pattern was tested for three iterations. Because 128 is greater than 100, the CRC32 path was selected. The observed CRC32 values were PATTERN=24650D57, UART5_RX=24650D57 and UART4_RX=24650D57, followed by UART CRC32 CHECK: PASS.

Edge cases

The implementation was tested with iterations=0, an unknown peripheral, invalid pattern length, and an oversized UDP command. Invalid commands returned the Failure result and did not start the requested peripheral test.

PC client

The formal PC implementation is a C++17 Linux CLI using UDP and SQLite. It supports run, all, history and get commands.

Hardware note

The assignment specifies STM32F746ZG, while the physically verified implementation uses NUCLEO-F756ZG / STM32F756ZGTx. The implementation was adapted to the actual target.